THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 49	$7 \times 7 = 49$
2	**(a)** 2^7	Add the exponents: $3 + 4 = 7$
3	**(a)** 17	Add the values of the symbols, subtracting a smaller symbol placed before a larger one
4	**(a)** 90°	The angles of a quadrilateral add up to 360°
5	**(c)** Yes	digit sum 15 is a multiple of 3
6	**(b)** 144	$12 \times 12 = 144$
7	**(c)** 2^5	Subtract the exponents: $9 - 4 = 5$
8	**(d)** VIII	8 is written as VIII
9	**(c)** 115°	Adjacent angles of a parallelogram add up to 180°
10	**(a)** Yes	digit sum 18 is a multiple of 9
11	**(b)** 13	$13 \times 13 = 169$
12	**(a)** 2^6	Multiply the exponents
13	**(b)** 5	Add the powers of 2 where there is a 1
14	**(d)** 58°	Opposite angles of a parallelogram are equal
15	**(d)** 3846	last digit 6 is even

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$27^2 = 27 \times 27 = 729$ (2).	2
17	Add exponents: 2^5 (1); $= 32$ (1).	1+1
18	27 = XXVII (2).	2
19	Sum of angles $= 360^\circ$ (1); fourth angle $= 360 - 270 = 90^\circ$ (1).	1+1
20	Yes (1); digit sum 15 is a multiple of 3 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$196 = 2^2 \times 7^2$ (1); pairing equal factors (1); $\sqrt{196} = 14$ (1).	1+1+1
22	$2^5 \times 2^4 = 2^9$ (1); $2^9 \div 2^6 = 2^3$ (1); value 8 (1).	1+1+1
23	XLVII = 47 (1); CLXIV = 164 (1); add the symbol values, subtract when a smaller symbol comes before a larger one (1).	1+1+1
24	$\angle B = 180 - 62 = 118^\circ$ (1); $\angle C = \angle A = 62^\circ$ (1); $\angle D = \angle B = 118^\circ$ (1).	1+1+1
	OR: By 2: Yes, last digit 0 is even (1); by 5: Yes, last digit 0 is 0 or 5 (1); by 10: Yes, last digit 0 is 0 (1).	1+1+1
25	$512 = 2^9$ (1); grouping equal factors in triplets (1); $\sqrt[3]{512} = 8$ (1).	1+1+1
	OR: $4.2 \times 10^5 = 420000$ (1); $3.5 \times 10^{-3} = 0.0035$ (1); move the decimal point right for a positive power and left for a negative power (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$180 = 2^2 \times 3^2 \times 5$ (1); primes with odd power: multiplier $= 5$ (2); product $= 900$ (1); square root $= 30$ (1).	1+2+1+1
	OR: (i) 32 (1); (ii) $\frac{1}{9}$ (1); (iii) 1 (1); (iv) sum $= 32 + \frac{1}{9} + 1 = \frac{298}{9}$ (2).	1+1+1+2
27	12 = XII (1); 25 = XXV (1); 38 = XXXVIII (1); 46 = XLVI (1); 59 = LIX (1).	1+1+1+1+1
	OR: Fourth angle $= 95^\circ$ (2); exterior angles $180 - 70 = 110^\circ, 70^\circ, 95^\circ, 85^\circ$ (2); their sum $= 360^\circ$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 64	$4 \times 4 \times 4 = 64$
2	**(b)** $\frac{1}{16}$	$a^{-n} = \frac{1}{a^n} = \frac{1}{16}$
3	**(b)** 101	Divide repeatedly by 2 and read the remainders upwards
4	**(c)** 24 cm	$4 \times 6 = 24$
5	**(b)** 21	Add all the digits
6	**(a)** 4	$4^3 = 64$
7	**(d)** 3.2×10^3	One non-zero digit before the decimal point
8	**(c)** 600	Place value = digit x value of its position
9	**(a)** 26 cm	$2(8 + 5)$
10	**(b)** even	Odd + odd = even
11	**(d)** 25	Sum of first n odd numbers $= n^2 = 25$
12	**(b)** 3400	Move the decimal point right for a positive power and left for a negative power
13	**(a)** 0 to 1	Base 2 uses 2 digits starting from 0
14	**(c)** 10 cm	Diagonals of a rectangle are equal
15	**(c)** 2743	Write the digits in the opposite order

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Yes, $225 = 15 \times 15$ (1+1).	1+1
17	$2^{-6} = \frac{1}{2^6}$ (1) $= \frac{1}{64}$ (1).	1+1
18	$1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 7$ (1+1).	1+1
19	Opposite angle 48° (1); adjacent angles $180 - 48 = 132^\circ$ each (1).	1+1
20	Known digit sum $= 19$ (1); smallest digit $= 8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$4 \times 10^3 + 0 \times 10^2 + 8 \times 10^1 + 6 \times 10^0$ (3).	3
22	BD = AC = 16 cm (1); OA = half of AC = 8 cm (1); OB = 8 cm (1).	1+1+1
23	Digit sum of the other digits $= 7$ (1); need a total that is a multiple of 3 (1); digits: 2, 5, 8 (1).	1+1+1
24	$1 + 3 + 5 + 7 + 9 + 11 = 36$ (3).	3
	OR: $73000000 = 7.3 \times 10^7$ (1); $0.0000064 = 6.4 \times 10^{-6}$ (1); one non-zero digit before the decimal point (1).	1+1+1
25	$1 \times 5^2 + 2 \times 5^1 + 3 \times 5^0$ (1+1); $= 38$ (1).	1+1+1
	OR: Diagonals bisect each other (1); AC = 2 x 5 = 10 cm (1); BD = 2 x 8 = 16 cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	By 2: Yes, last digit 0 is even (1); By 3: Yes, digit sum 18 is a multiple of 3 (1); By 5: Yes, last digit 0 is 0 or 5 (1); By 9: Yes, digit sum 18 is a multiple of 9 (1); By 10: Yes, last digit 0 is 0 (1);	1+1+1+1+1
	OR: $108 = 2^2 \times 3^3$ (1); smallest divisor $= 4$ (2); quotient $= 27$ (1); cube root $= 3$ (1).	1+2+1+1
27	$(2^2)^3 = 2^6$ (1); divide: 2^2 (1); multiply: 2^3 (2); value 8 (1).	1+1+2+1
	OR: Steps: $19 = 2 \times 9 + 1$; $9 = 2 \times 4 + 1$; $4 = 2 \times 2 + 0$; $2 = 2 \times 1 + 0$; $1 = 2 \times 0 + 1$ (3); reading remainders from last to first gives 10011_2 (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 64	$8 \times 8 = 64$
2	**(a)** 3^7	Add the exponents: $2 + 5 = 7$
3	**(c)** L	Learn the seven Roman symbols
4	**(a)** 8 cm	Diagonals of a parallelogram bisect each other
5	**(a)** 4554	A palindrome reads the same from both ends
6	**(c)** 196	$14 \times 14 = 196$
7	**(c)** 3^5	Subtract the exponents: $8 - 3 = 5$
8	**(c)** 24	Add the values of the symbols, subtracting a smaller symbol placed before a larger one
9	**(d)** 4 sides and 4 vertices	Quad means four
10	**(d)** 12	Smallest common multiple
11	**(c)** 17	$17 \times 17 = 289$
12	**(b)** 3^8	Multiply the exponents
13	**(a)** XIX	19 is written as XIX
14	**(d)** 100°	The angles of a quadrilateral add up to 360°
15	**(d)** No	digit sum 17 is not a multiple of 3

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$13^3 = 13 \times 13 \times 13 = 2197$ (2).	2
17	2.8×10^4 (2).	2
18	Repeated division by 2 gives 7 = 111_2 (1+1).	1+1
19	Square (1); diagonals are equal and perpendicular (1).	1+1
20	Multiples of 3: 3, 6, 9, 12 (1); LCM $= 12$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	12 and 16: common factors give HCF $= 4$ (1.5); LCM $= 48$ (1.5).	1½+1½
22	$324 = 2^2 \times 3^4$ (1); pairing equal factors (1); $\sqrt{324} = 18$ (1).	1+1+1
23	$3^4 \times 3^3 = 3^7$ (1); $3^7 \div 3^5 = 3^2$ (1); value 9 (1).	1+1+1
24	XXXVIII = 38 (1); CCXXIX = 229 (1); add the symbol values, subtract when a smaller symbol comes before a larger one (1).	1+1+1
	OR: $\angle B = 180 - 78 = 102^\circ$ (1); $\angle C = \angle A = 78^\circ$ (1); $\angle D = \angle B = 102^\circ$ (1).	1+1+1
25	By 2: No, last digit 5 is not even (1); by 5: Yes, last digit 5 is 0 or 5 (1); by 10: No, last digit 5 is not 0 (1).	1+1+1
	OR: $729 = 3^6$ (1); grouping equal factors in triplets (1); $\sqrt[3]{729} = 9$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$\angle B = 114^\circ$ (1); $\angle C = \angle A = 66^\circ$ (1); $\angle D = \angle B = 114^\circ$ (1); perimeter $= 4 \times 7 = 28$ cm (2).	1+1+1+2
	OR: Remainder when 4537 is divided by 9 is 1 (2); number to add $= 9 - 1 = 8$ (2); new number 4545 (1).	2+2+1
27	$252 = 2^2 \times 3^2 \times 7$ (1); primes with odd power: multiplier $= 7$ (2); product $= 1764$ (1); square root $= 42$ (1).	1+2+1+1
	OR: (i) 27 (1); (ii) $\frac{1}{16}$ (1); (iii) 1 (1); (iv) sum $= 27 + \frac{1}{16} + 1 = \frac{449}{16}$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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